

## TAICHUNG INTL, Taiwan

WMO: 467700

Lat: 24.265N

Lon: 120.621E

Elev: 202

StdP: 98.92

Time Zone: 8.00 (E08)

Period: 90-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 8.1        | 9.2 | 3.0                           | 4.8 | 9.5  | 5.0 | 5.5 | 10.7 | 13.9                  | 9.2  | 12.4 | 10.7 | 7.2                   | 20   | 0.404 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 6.8                    | 32.8            | 26.7 | 32.1 | 26.4 | 31.8 | 26.4 | 27.9                | 30.9 | 27.3 | 30.6 | 26.9 | 30.3 | 4.8                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.1                            | 23.4 | 30.1 | 26.2 | 22.2 | 29.5 | 26.0 | 22.0 | 29.3 | 90.8          | 30.8 | 88.0 | 30.9 | 86.3 | 30.5 | 30.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 12.2              | 10.5 | 9.4 | 5.4                        | 34.5 | 3.2                | 1.2 | 3.1                                                | 35.3 | 1.2        | 36.0 | -0.6       | 36.7 | -3.0       | 37.6 |  |  |
|                   |      |     | 3.6                        | 29.0 | 2.2                | 0.8 | 2.0                                                | 29.6 | 0.7        | 30.0 | -0.5       | 30.4 | -2.1       | 31.0 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 22.5   | 16.0  | 16.0  | 18.1  | 22.1  | 25.0  | 27.2  | 28.2  | 27.7  | 26.7  | 24.1  | 21.3  | 17.8 |
|                                                                                  | DBStd       | 5.07   | 2.80  | 3.33  | 3.46  | 3.11  | 2.38  | 1.64  | 1.12  | 1.27  | 1.74  | 2.09  | 2.56  | 2.78 |
|                                                                                  | HDD10.0     | 3      | 2     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1    |
|                                                                                  | HDD18.3     | 267    | 82    | 80    | 48    | 8     | 1     | 0     | 0     | 0     | 0     | 0     | 6     | 43   |
|                                                                                  | CDD10.0     | 4581   | 186   | 168   | 253   | 362   | 464   | 514   | 566   | 549   | 500   | 437   | 339   | 242  |
|                                                                                  | CDD18.3     | 1803   | 8     | 14    | 42    | 119   | 207   | 264   | 307   | 291   | 250   | 179   | 96    | 26   |
|                                                                                  | CDH23.3     | 15016  | 16    | 43    | 161   | 606   | 1547  | 2567  | 3345  | 2993  | 2220  | 1092  | 367   | 59   |
| CDH26.7                                                                          | 4669        | 1      | 4     | 19    | 109   | 405   | 849   | 1254  | 1013  | 690   | 266   | 56    | 4     |      |
| Wind                                                                             | WSAvg       | 4.2    | 5.2   | 4.9   | 4.2   | 3.6   | 3.1   | 3.6   | 3.5   | 3.3   | 3.5   | 4.8   | 4.9   | 5.4  |
| Precipitation                                                                    | PrecAvg     | 1506   | 34    | 84    | 97    | 144   | 195   | 268   | 218   | 238   | 144   | 31    | 25    | 30   |
|                                                                                  | PrecMax     | 2246   | 107   | 337   | 206   | 517   | 481   | 669   | 582   | 594   | 721   | 232   | 167   | 84   |
|                                                                                  | PrecMin     | 696    | 2     | 3     | 30    | 6     | 23    | 23    | 35    | 25    | 12    | 1     | 0     | 1    |
|                                                                                  | PrecStd     | 419    | 23    | 81    | 53    | 114   | 124   | 160   | 149   | 163   | 186   | 51    | 38    | 25   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 25.1  | 26.9  | 28.8  | 30.8  | 32.1  | 32.9  | 33.8  | 33.1  | 33.1  | 32.2  | 29.9  | 27.1 |
|                                                                                  |             | MCWB   | 19.1  | 21.0  | 22.8  | 25.3  | 25.9  | 26.6  | 27.2  | 27.2  | 26.5  | 25.4  | 23.7  | 20.5 |
|                                                                                  | 2%          | DB     | 23.1  | 24.5  | 26.8  | 29.1  | 31.0  | 32.1  | 32.9  | 32.2  | 32.0  | 30.8  | 28.1  | 25.0 |
|                                                                                  |             | MCWB   | 18.3  | 20.0  | 21.9  | 24.5  | 25.9  | 26.4  | 26.7  | 26.8  | 26.1  | 25.4  | 23.0  | 20.4 |
|                                                                                  | 5%          | DB     | 21.5  | 22.9  | 25.2  | 28.0  | 30.0  | 31.3  | 32.1  | 31.8  | 31.1  | 29.1  | 27.0  | 23.2 |
|                                                                                  |             | MCWB   | 17.5  | 19.1  | 20.8  | 23.8  | 25.7  | 26.3  | 26.5  | 26.7  | 26.1  | 24.2  | 22.0  | 19.1 |
|                                                                                  | 10%         | DB     | 20.0  | 21.0  | 23.2  | 26.8  | 29.0  | 30.9  | 31.7  | 31.0  | 30.1  | 28.1  | 25.8  | 22.0 |
|                                                                                  |             | MCWB   | 16.7  | 17.8  | 19.9  | 23.1  | 25.1  | 26.3  | 26.5  | 26.7  | 25.9  | 23.5  | 21.3  | 18.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 20.5  | 22.6  | 23.8  | 26.1  | 27.2  | 28.0  | 28.7  | 28.8  | 27.9  | 26.8  | 24.9  | 22.2 |
|                                                                                  |             | MCDB   | 23.2  | 25.2  | 27.5  | 29.0  | 30.2  | 30.5  | 31.5  | 31.6  | 30.8  | 30.1  | 28.5  | 25.6 |
|                                                                                  | 2%          | WB     | 19.3  | 20.8  | 22.6  | 25.1  | 26.5  | 27.3  | 27.8  | 27.8  | 27.1  | 26.0  | 23.8  | 20.8 |
|                                                                                  |             | MCDB   | 22.2  | 23.4  | 25.7  | 28.1  | 29.7  | 30.5  | 31.1  | 30.9  | 30.4  | 29.1  | 27.2  | 23.8 |
|                                                                                  | 5%          | WB     | 18.4  | 19.5  | 21.6  | 24.3  | 25.9  | 26.8  | 27.2  | 27.2  | 26.6  | 25.4  | 22.9  | 19.9 |
|                                                                                  |             | MCDB   | 20.9  | 21.9  | 24.3  | 27.1  | 28.9  | 30.2  | 30.9  | 30.4  | 29.7  | 28.3  | 25.9  | 22.5 |
|                                                                                  | 10%         | WB     | 17.4  | 18.3  | 20.5  | 23.6  | 25.4  | 26.4  | 26.7  | 26.7  | 26.1  | 24.6  | 22.0  | 18.9 |
|                                                                                  |             | MCDB   | 19.5  | 20.8  | 22.8  | 26.2  | 28.3  | 29.6  | 30.3  | 29.8  | 29.1  | 27.3  | 24.5  | 21.3 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.5   | 7.7   | 7.6   | 7.1   | 6.5   | 6.3   | 6.8   | 6.4   | 6.8   | 7.7   | 7.9   | 7.8  |
|                                                                                  |             | MCDBR  | 9.1   | 10.3  | 10.0  | 8.2   | 7.2   | 7.1   | 7.5   | 7.3   | 7.4   | 8.3   | 9.2   | 9.2  |
|                                                                                  |             | MCWBR  | 5.5   | 6.4   | 5.9   | 5.0   | 4.1   | 3.6   | 3.7   | 3.6   | 3.7   | 4.6   | 5.0   | 5.7  |
|                                                                                  | 5% WB       | MCDBR  | 8.2   | 10.1  | 9.6   | 7.5   | 6.7   | 6.7   | 7.2   | 6.8   | 6.8   | 7.5   | 8.5   | 8.9  |
|                                                                                  |             | MCWBR  | 5.5   | 6.8   | 6.2   | 5.0   | 4.3   | 4.0   | 4.0   | 4.0   | 4.0   | 4.6   | 5.2   | 5.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.486  | 0.540 | 0.656 | 0.666 | 0.570 | 0.524 | 0.537 | 0.568 | 0.558 | 0.522 | 0.493 | 0.486 |      |
|                                                                                  | taud        | 2.017  | 1.902 | 1.667 | 1.699 | 1.969 | 2.118 | 2.073 | 1.976 | 1.977 | 2.021 | 2.081 | 2.034 |      |
|                                                                                  | Ebn at Noon | 757    | 744   | 681   | 687   | 754   | 783   | 773   | 750   | 747   | 751   | 746   | 737   |      |
|                                                                                  | Edh at Noon | 156    | 187   | 248   | 245   | 186   | 159   | 166   | 183   | 179   | 164   | 146   | 149   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.87   | 3.17  | 3.68  | 4.06  | 4.68  | 5.16  | 5.89  | 5.34  | 4.89  | 4.31  | 3.22  | 2.85  |      |
|                                                                                  | RadStd      | 0.40   | 0.47  | 0.41  | 0.55  | 0.71  | 0.56  | 0.36  | 0.52  | 0.47  | 0.37  | 0.31  | 0.35  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (28 neighbors) | +0.25 | N/A     | N/A    | +0.22   | N/A   | N/A   | N/A         | N/A     | +127    | +112    |

Nomenclature: See separate page